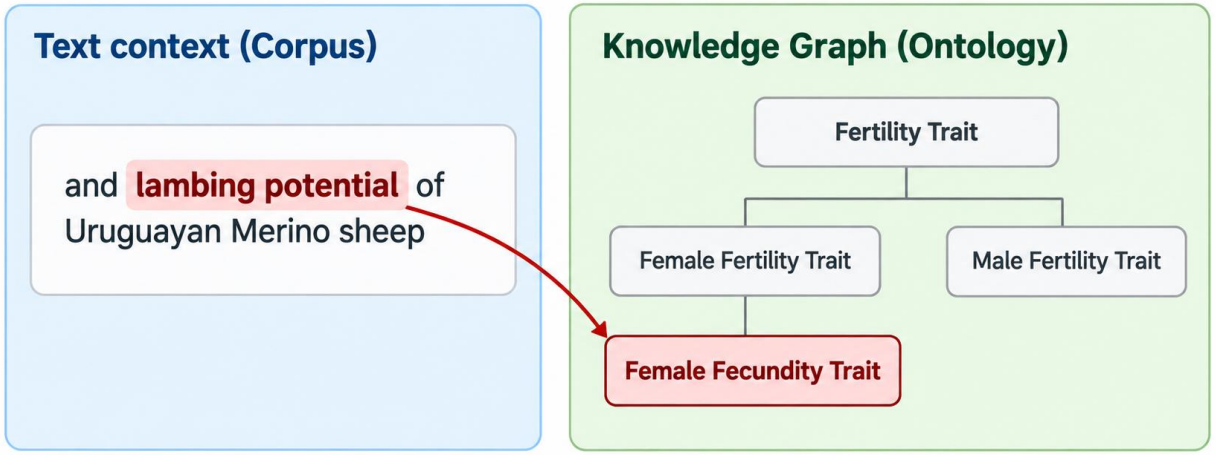
Bridging Lexical Divergence: LLM-Assisted, Cost-Efficient, Zero-shot Scientific Entity Linking

IOWA STATE UNIVERSITY
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Presenter: Md Rasel Khondokar

Introduction



- ❑ Lexical challenge : Scientific-domain vs. General-domain
- ❑ Handling specialized terminology
- ❑ Ignored ontology structure
- ❑ Natural remedy : In-domain fine-tuning
  - Expensive expert annotation

Per-Mention to Entity Alias Augmentation

- Per-Mention Augmentation
  - Token cost scales with corpus size
  - Tied to corpus and not reusable

- Entity Alias Augmentation
  - Generate aliases for a small subset of entities

- **Contributions**
  - Animal science benchmark
  - Sci-ZSEL
  - Empirical study

A New Dataset in Animal Science Domain

- Trait names differ across aspects, species
- A mention can link to different ontologies
  - Clinical Measurement Ontology (CMO)
  - Vertebrate Trait Ontology (VT)
  - Livestock Product Trait Ontology (LPT)
- Train dataset
  - Training corpus: PubMed ID from AnimalQTLdb (Hu et al. ,2007)
  - Trait mentions recognition : CuPUL (Li et al., 2025)
- Test dataset
  - Expert curated

Animal Science Benchmark

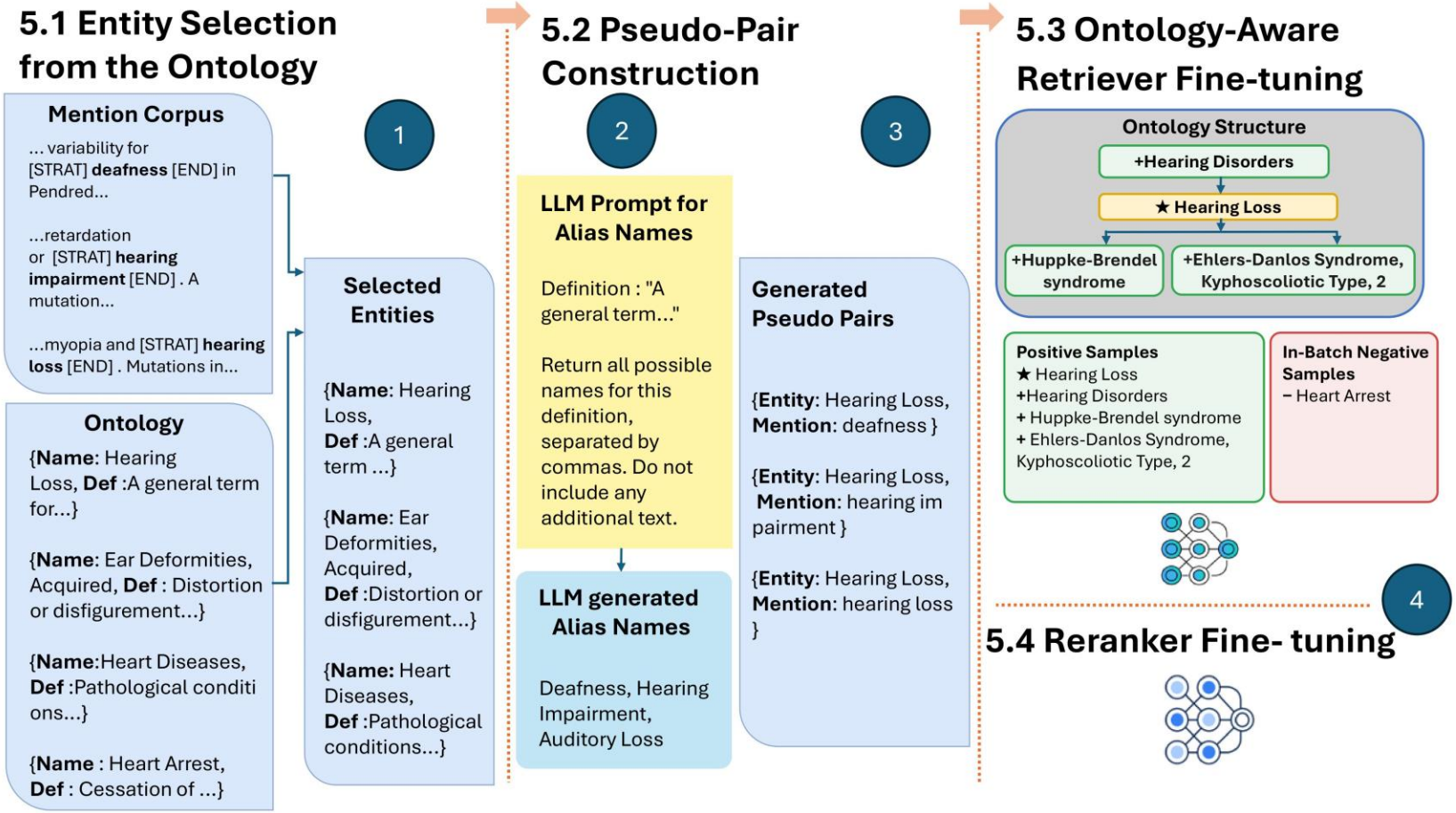
	#Samples	#Ment	#Ent	NO%	Syn/Ent
QTL_{CMO}	2032	458	107	43.45	1.07
QTL_{VT}	1688	500	111	57.11	0.91
QTL_{LPT}	722	233	103	45.01	0.89
NCBI	960	287	190	22.92	9.57
BC5CDR	9465	2084	1267	24.16	2.13

- ❑ Higher No Overlap (NO) rate
- ❑ Fewer synonyms per entity
- ❑ Significantly harder than the existing benchmark

Data & Code



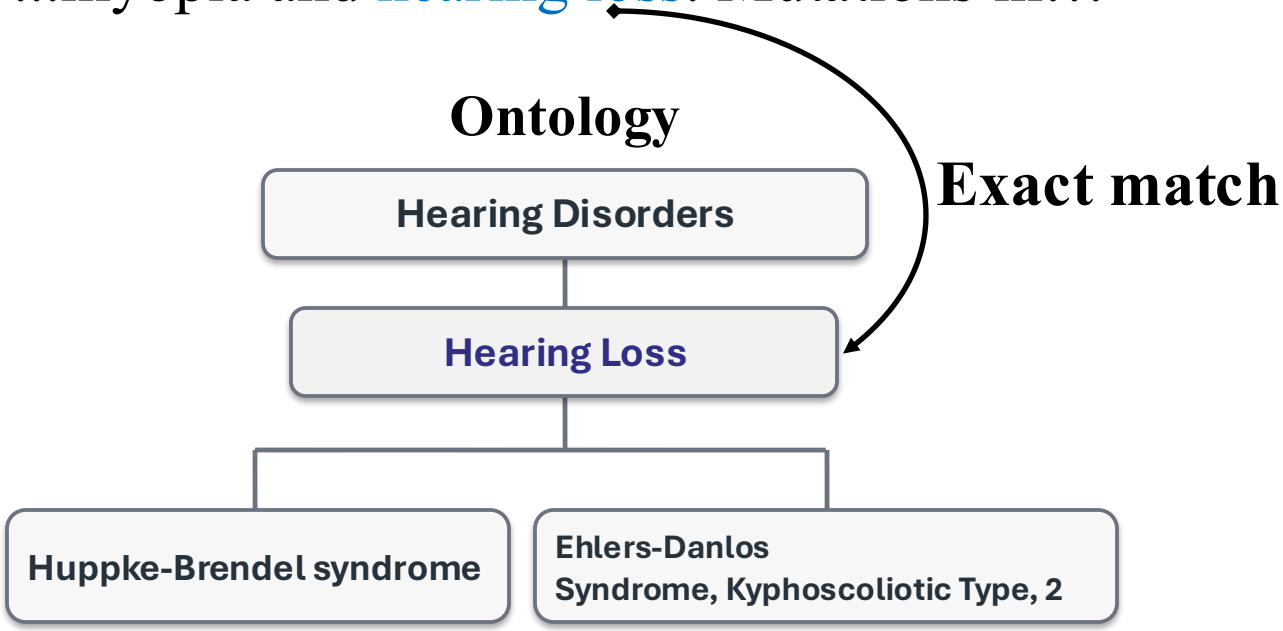
Sci-ZSEL Overview



Entity Selection from the Ontology - EM

Mention Corpus

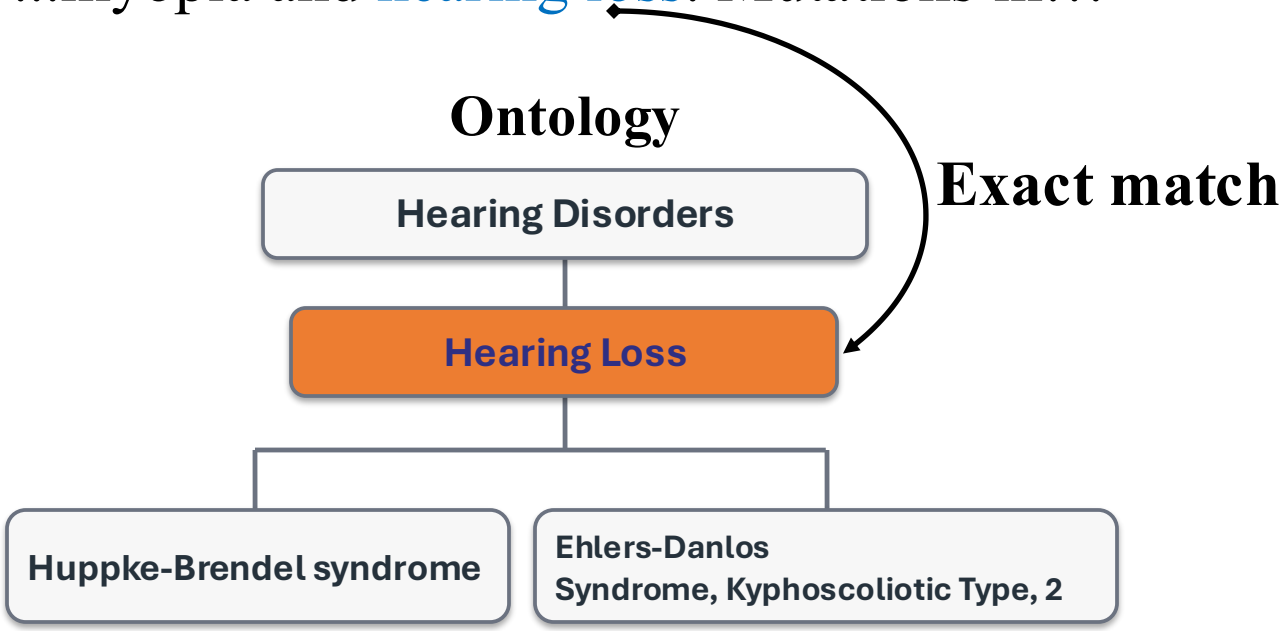
...myopia and hearing loss. Mutations in…



Entity Selection from the Ontology - EM

Mention Corpus

...myopia and hearing loss. Mutations in…



Entity Selection from the Ontology - BT

Mention Corpus

...retardation or hearing impairment. A mutation…

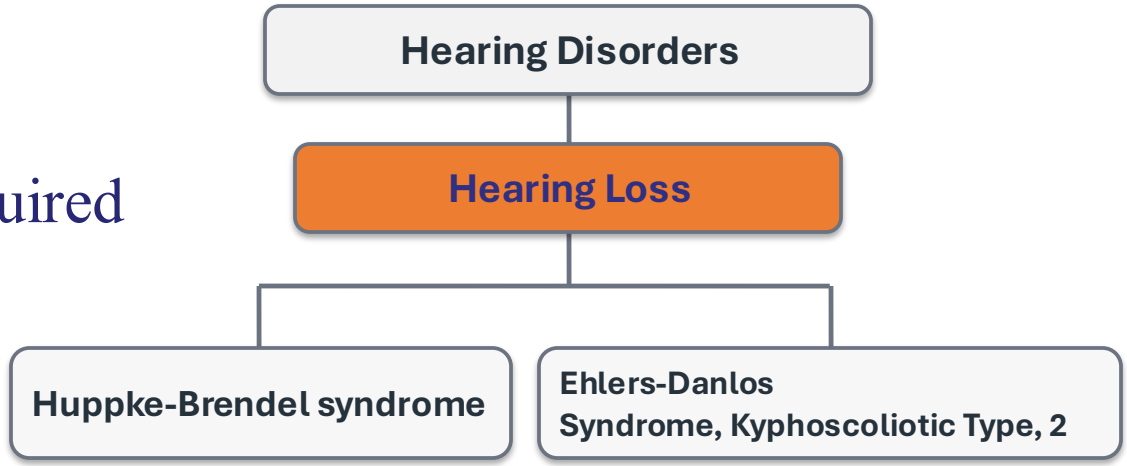
Bi-encoder ranking

1. Hearing disorders
2. Ear Deformities, Acquired

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Ontology



Entity Selection from the Ontology - BT

Mention Corpus

...retardation or hearing impairment. A mutation…

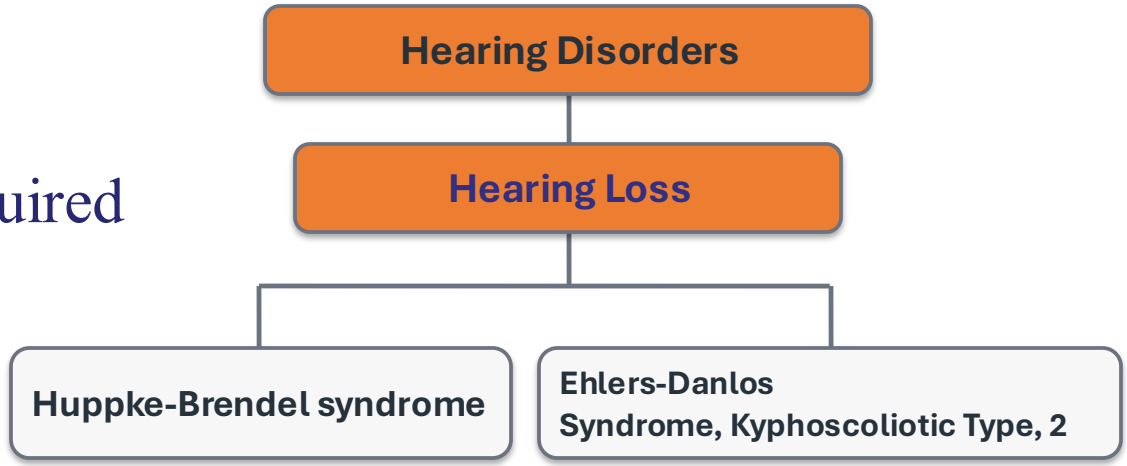
Bi-encoder ranking

1. Hearing disorders
2. Ear Deformities, Acquired

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Ontology



Pseudo-Pair Construction : LLM-generated Alias

Hearing Loss: "A general term for the complete or partial loss of the ability to hear from one or both ears"

Pseudo-Pair Construction : LLM-generated Alias

LLM A general term for the complete or partial loss of the ability to hear from one or both ears

Pseudo-Pair Construction : LLM-generated Alias

LLM A general term for the complete or partial loss of the ability to hear from one or both ears

Generate Alias

- Deafness, Hearing Impairment, Auditory Loss

Pseudo-Pair Construction : LLM-generated Alias

LLM A general term for the complete or partial loss of the ability to hear from one or both ears

Generate Alias

- Deafness, Hearing Impairment, Auditory Loss



Mention Corpus

...variability for deafness in Pendred…

Pseudo-Pair Construction

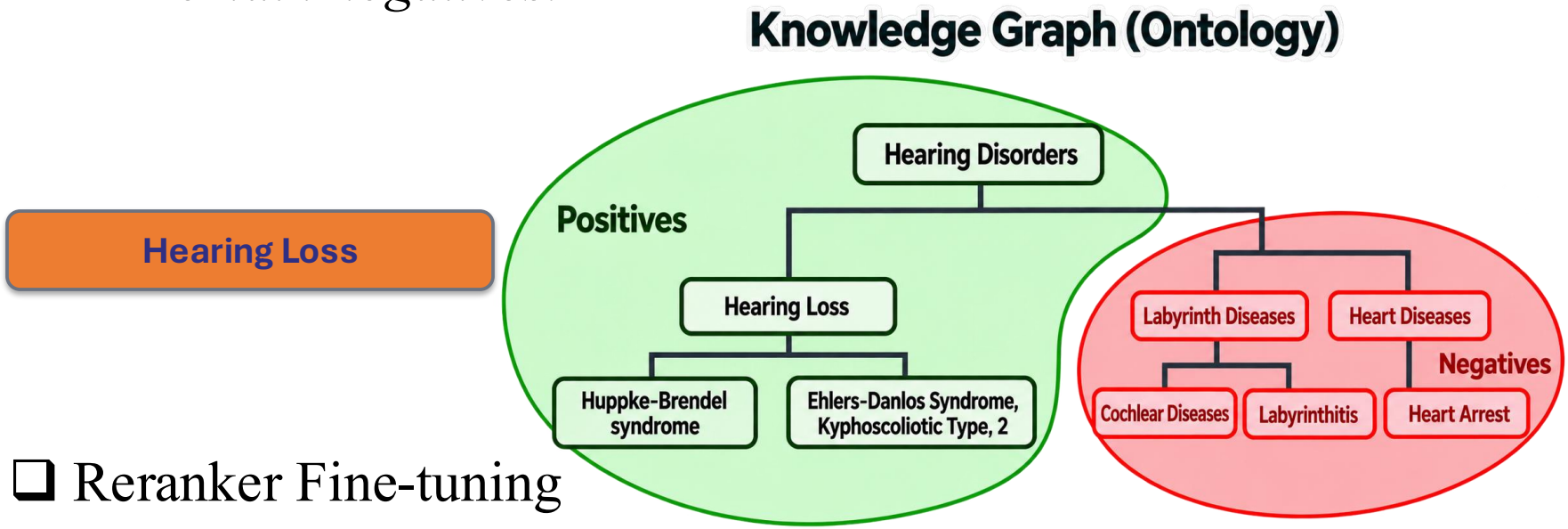
Pseudo Pairs

{hearing loss, Hearing Loss}	Exact match
{deafness, Hearing Loss}	LLM augmented match
{cancer, Neoplasms}	Synonym match

….

Retriever and reranker fine-tuning

- Ontology-Aware Retriever Fine-tuning
  - ***PC-POS***: *Parents and children added as positives; siblings remain negatives.*



- Reranker Fine-tuning
  - Positive: Pseudo-Pair
  - Negative: other top-K pairs from retriever

Experimental setup

- Backbones
  - Retriever: BLINK
  - Reranker: BLINK, ReS
- Evaluation metrics
  - MRR
  - Recall@64 for Retriever, Recall@1 for Reranker
  - Category-wise evaluation : Mention vs Entity name
    - High Overlap (HO): Substring
    - Low Overlap (LO): Partial lexical overlap
    - No Overlap (NO): No lexical overlap

Retriever Results

Dataset	Setting	BASE	RM-PCS	PC-POS
NCBI	No fine-tune	$88.12_{\pm 0.00}$	$88.12_{\pm 0.00}$	$88.12_{\pm 0.00}$
	Sci-ZSEL w/o filter	$87.74_{\pm 0.22}$	$87.01_{\pm 1.85}$	$\mathbf{88.75}_{\pm 0.36}$
	Sci-ZSEL	$87.40_{\pm 0.47}$	$80.70_{\pm 11.97}$	$\mathbf{87.81}_{\pm 0.63}$
	Synonym	$89.20_{\pm 0.24}$	$89.73_{\pm 0.30}$	$\mathbf{90.21}_{\pm 0.36}$
	Sci-ZSEL + Synonym	$89.72_{\pm 0.97}$	$89.41_{\pm 0.78}$	$\mathbf{90.80}_{\pm 0.87}$
BC5CDR	No fine-tune	$88.79_{\pm 0.00}$	$88.79_{\pm 0.00}$	$88.79_{\pm 0.00}$
	Sci-ZSEL w/o filter	$88.41_{\pm 0.74}$	$88.21_{\pm 2.01}$	$\mathbf{88.93}_{\pm 1.03}$
	Sci-ZSEL	$85.27_{\pm 1.75}$	$85.41_{\pm 0.27}$	$\mathbf{87.80}_{\pm 0.74}$
	Synonym	$87.22_{\pm 0.16}$	$87.13_{\pm 0.59}$	$\mathbf{88.57}_{\pm 0.50}$
	Sci-ZSEL + Synonym	$87.96_{\pm 0.48}$	$86.61_{\pm 2.16}$	$\mathbf{89.14}_{\pm 0.65}$
QTL_{CMO}	No fine-tune	$75.74_{\pm 0.00}$	$75.74_{\pm 0.00}$	$75.74_{\pm 0.00}$
	Sci-ZSEL w/o filter	$\mathbf{89.96}_{\pm 0.23}$	$88.02_{\pm 1.24}$	$89.24_{\pm 1.61}$
	Sci-ZSEL	$\mathbf{89.22}_{\pm 0.86}$	$88.06_{\pm 0.25}$	$88.34_{\pm 1.37}$
	Synonym	$\mathbf{85.71}_{\pm 0.40}$	$85.47_{\pm 0.49}$	$85.45_{\pm 0.91}$
	Sci-ZSEL + Synonym	$88.22_{\pm 0.89}$	$89.03_{\pm 1.68}$	$\mathbf{89.09}_{\pm 1.57}$
QTL_{VT}	No fine-tune	$72.63_{\pm 0.00}$	$72.63_{\pm 0.00}$	$72.63_{\pm 0.00}$
	Sci-ZSEL w/o filter	$\mathbf{84.70}_{\pm 0.96}$	$84.52_{\pm 1.08}$	$83.69_{\pm 0.94}$
	Sci-ZSEL	$84.72_{\pm 1.69}$	$\mathbf{85.51}_{\pm 0.30}$	$85.31_{\pm 1.03}$
	Synonym	$86.99_{\pm 1.74}$	$86.79_{\pm 2.67}$	$\mathbf{87.91}_{\pm 2.22}$
	Sci-ZSEL + Synonym	$85.58_{\pm 4.10}$	$\mathbf{88.69}_{\pm 2.22}$	$87.99_{\pm 1.47}$
QTL_{LPT}	No fine-tune	$68.70_{\pm 0.00}$	$68.70_{\pm 0.00}$	$68.70_{\pm 0.00}$
	Sci-ZSEL w/o filter	$70.08_{\pm 0.28}$	$70.82_{\pm 0.97}$	$\mathbf{71.01}_{\pm 0.92}$
	Sci-ZSEL	$75.07_{\pm 1.54}$	$\mathbf{76.87}_{\pm 0.73}$	$76.18_{\pm 0.37}$
	Synonym	$\mathbf{92.43}_{\pm 0.68}$	$91.83_{\pm 0.42}$	$91.78_{\pm 1.24}$
	Sci-ZSEL + Synonym	$92.29_{\pm 0.44}$	$92.48_{\pm 0.35}$	$\mathbf{92.66}_{\pm 0.69}$

Fine-tuning improves on all benchmarks, though depends on synonym density. Gains are much larger on the synonym-sparse animal-science benchmarks.

Reranker Results

Dataset	Setting	BLINK					ReS				
		Overall	MRR	HO	LO	NO	Overall	MRR	HO	LO	NO
NCBI	No fine-tune	64.38$_{\pm 0.00}$	72.39$_{\pm 0.00}$	**93.83**$_{\pm 0.00}$	64.12$_{\pm 0.00}$	23.64$_{\pm 0.00}$	45.94$_{\pm 0.00}$	58.41$_{\pm 0.00}$	67.53$_{\pm 0.00}$	44.44$_{\pm 0.00}$	18.64$_{\pm 0.00}$
	Sci-ZSEL w/o filter	70.69$_{\pm 0.58}$	77.65$_{\pm 0.47}$	92.75$_{\pm 0.19}$	66.44$_{\pm 1.22}$	48.18$_{\pm 0.79}$	60.28$_{\pm 0.42}$	73.57$_{\pm 0.27}$	68.72$_{\pm 0.68}$	59.41$_{\pm 1.88}$	50.15$_{\pm 3.19}$
	Sci-ZSEL	71.88$_{\pm 0.42}$	78.48$_{\pm 0.35}$	92.86$_{\pm 0.00}$	68.90$_{\pm 1.05}$	48.33$_{\pm 0.27}$	63.06$_{\pm 2.25}$	73.00$_{\pm 2.50}$	73.05$_{\pm 1.17}$	68.52$_{\pm 2.34}$	38.33$_{\pm 6.45}$
	Synonym	76.18$_{\pm 0.24}$	83.17$_{\pm 0.49}$	93.18$_{\pm 0.33}$	73.23$_{\pm 0.58}$	58.18$_{\pm 2.08}$	**78.09**$_{\pm 0.73}$	**84.59**$_{\pm 0.37}$	**93.94**$_{\pm 2.40}$	**74.07**$_{\pm 0.70}$	**63.79**$_{\pm 1.89}$
	Sci-ZSEL + Synonym	**77.43**$_{\pm 0.67}$	**84.35**$_{\pm 0.74}$	92.75$_{\pm 1.50}$	**74.54**$_{\pm 2.06}$	**61.67**$_{\pm 1.15}$	70.52$_{\pm 1.02}$	79.11$_{\pm 1.58}$	74.24$_{\pm 1.46}$	71.30$_{\pm 0.24}$	**63.79**$_{\pm 3.09}$
BC5CDR	No fine-tune	74.36$_{\pm 0.00}$	79.53$_{\pm 0.00}$	**96.26**$_{\pm 0.00}$	66.77$_{\pm 0.00}$	22.43$_{\pm 0.00}$	32.65$_{\pm 0.00}$	45.34$_{\pm 0.00}$	39.86$_{\pm 0.00}$	35.61$_{\pm 0.00}$	12.42$_{\pm 0.00}$
	Sci-ZSEL w/o filter	78.14$_{\pm 0.47}$	83.90$_{\pm 0.36}$	94.94$_{\pm 0.64}$	61.55$_{\pm 1.31}$	44.45$_{\pm 1.04}$	68.14$_{\pm 0.76}$	77.01$_{\pm 0.67}$	82.77$_{\pm 0.53}$	51.04$_{\pm 2.87}$	40.31$_{\pm 1.32}$
	Sci-ZSEL	79.16$_{\pm 0.58}$	84.40$_{\pm 0.27}$	96.01$_{\pm 0.31}$	72.17$_{\pm 2.03}$	39.88$_{\pm 1.45}$	76.91$_{\pm 0.53}$	82.52$_{\pm 0.28}$	**94.83**$_{\pm 0.58}$	64.65$_{\pm 1.58}$	37.85$_{\pm 1.42}$
	Synonym	78.20$_{\pm 1.24}$	83.49$_{\pm 1.88}$	96.17$_{\pm 0.24}$	67.97$_{\pm 1.93}$	37.85$_{\pm 3.51}$	74.74$_{\pm 0.40}$	80.15$_{\pm 0.48}$	94.12$_{\pm 0.44}$	58.68$_{\pm 3.04}$	34.09$_{\pm 0.83}$
	Sci-ZSEL + Synonym	**81.20**$_{\pm 0.58}$	**86.20**$_{\pm 0.48}$	95.96$_{\pm 0.30}$	**74.93**$_{\pm 1.96}$	**46.87**$_{\pm 1.30}$	**77.90**$_{\pm 0.12}$	**83.21**$_{\pm 0.26}$	94.78$_{\pm 0.17}$	**65.42**$_{\pm 2.49}$	**41.66**$_{\pm 0.66}$
QTL_{CMO}	No fine-tune	50.15$_{\pm 0.00}$	60.74$_{\pm 0.00}$	92.91$_{\pm 0.00}$	50.95$_{\pm 0.00}$	29.67$_{\pm 0.00}$	48.92$_{\pm 0.00}$	61.06$_{\pm 0.00}$	82.15$_{\pm 0.00}$	51.49$_{\pm 0.00}$	31.37$_{\pm 0.00}$
	Sci-ZSEL w/o filter	57.09$_{\pm 2.37}$	66.43$_{\pm 1.96}$	93.97$_{\pm 3.88}$	58.74$_{\pm 2.26}$	38.62$_{\pm 2.31}$	41.42$_{\pm 2.31}$	54.82$_{\pm 2.24}$	44.34$_{\pm 2.98}$	49.59$_{\pm 4.06}$	33.22$_{\pm 4.05}$
	Sci-ZSEL	56.89$_{\pm 0.48}$	67.15$_{\pm 0.34}$	93.16$_{\pm 0.42}$	56.89$_{\pm 0.14}$	**40.09**$_{\pm 1.08}$	56.84$_{\pm 1.46}$	66.72$_{\pm 1.46}$	96.74$_{\pm 0.79}$	58.43$_{\pm 1.97}$	**37.03**$_{\pm 2.98}$
	Synonym	57.59$_{\pm 1.72}$	66.50$_{\pm 1.51}$	93.15$_{\pm 0.25}$	62.79$_{\pm 2.24}$	36.77$_{\pm 2.34}$	**58.48**$_{\pm 0.37}$	**68.29**$_{\pm 0.36}$	94.70$_{\pm 0.51}$	64.91$_{\pm 0.41}$	36.32$_{\pm 1.09}$
	Sci-ZSEL + Synonym	**58.88**$_{\pm 3.30}$	**67.72**$_{\pm 2.92}$	**94.87**$_{\pm 0.65}$	**65.22**$_{\pm 2.99}$	36.88$_{\pm 5.58}$	57.43$_{\pm 0.23}$	66.95$_{\pm 0.12}$	**96.99**$_{\pm 0.79}$	**65.54**$_{\pm 0.36}$	32.31$_{\pm 0.91}$
QTL_{VT}	No fine-tune	43.13$_{\pm 0.00}$	56.38$_{\pm 0.00}$	97.32$_{\pm 0.00}$	48.70$_{\pm 0.00}$	31.43$_{\pm 0.00}$	38.33$_{\pm 0.00}$	51.14$_{\pm 0.00}$	95.97$_{\pm 0.00}$	54.61$_{\pm 0.00}$	19.71$_{\pm 0.00}$
	Sci-ZSEL w/o filter	38.29$_{\pm 1.21}$	54.51$_{\pm 0.74}$	97.32$_{\pm 0.00}$	50.90$_{\pm 2.58}$	21.64$_{\pm 0.94}$	45.81$_{\pm 2.13}$	59.78$_{\pm 0.86}$	97.32$_{\pm 0.00}$	64.12$_{\pm 5.84}$	26.94$_{\pm 0.40}$
	Sci-ZSEL	46.64$_{\pm 0.54}$	60.04$_{\pm 0.26}$	97.32$_{\pm 0.00}$	49.68$_{\pm 0.96}$	37.00$_{\pm 0.43}$	41.15$_{\pm 0.42}$	53.98$_{\pm 0.38}$	97.32$_{\pm 0.00}$	59.88$_{\pm 1.50}$	21.30$_{\pm 0.76}$
	Synonym	70.83$_{\pm 0.42}$	**79.66**$_{\pm 0.73}$	97.54$_{\pm 0.39}$	**76.64**$_{\pm 1.40}$	63.24$_{\pm 1.32}$	62.78$_{\pm 3.91}$	73.65$_{\pm 3.55}$	96.87$_{\pm 0.39}$	67.54$_{\pm 0.61}$	54.67$_{\pm 6.58}$
	Sci-ZSEL + Synonym	**71.25**$_{\pm 2.58}$	79.49$_{\pm 1.65}$	**97.77**$_{\pm 0.77}$	76.46$_{\pm 2.47}$	**64.04**$_{\pm 3.43}$	**65.58**$_{\pm 2.62}$	**75.76**$_{\pm 2.03}$	**97.77**$_{\pm 0.77}$	**74.09**$_{\pm 4.31}$	**55.53**$_{\pm 2.90}$
QTL_{LPT}	No fine-tune	57.34$_{\pm 0.00}$	63.29$_{\pm 0.00}$	94.59$_{\pm 0.00}$	78.29$_{\pm 0.00}$	20.62$_{\pm 0.00}$	59.70$_{\pm 0.00}$	67.91$_{\pm 0.00}$	92.34$_{\pm 0.00}$	77.71$_{\pm 0.00}$	27.69$_{\pm 0.00}$
	Sci-ZSEL w/o filter	54.62$_{\pm 1.06}$	61.67$_{\pm 0.85}$	95.05$_{\pm 0.46}$	72.00$_{\pm 2.29}$	17.64$_{\pm 1.45}$	55.82$_{\pm 0.64}$	64.20$_{\pm 0.84}$	93.54$_{\pm 2.03}$	68.95$_{\pm 1.44}$	22.97$_{\pm 0.89}$
	Sci-ZSEL	58.22$_{\pm 0.29}$	64.18$_{\pm 0.19}$	95.35$_{\pm 0.52}$	**79.81**$_{\pm 1.43}$	21.23$_{\pm 0.62}$	62.14$_{\pm 0.81}$	69.71$_{\pm 0.60}$	94.29$_{\pm 0.26}$	80.38$_{\pm 1.32}$	30.36$_{\pm 0.99}$
	Synonym	78.81$_{\pm 1.92}$	84.56$_{\pm 1.60}$	97.90$_{\pm 0.26}$	79.43$_{\pm 0.57}$	65.44$_{\pm 4.20}$	77.93$_{\pm 1.53}$	83.75$_{\pm 1.14}$	95.05$_{\pm 0.46}$	77.14$_{\pm 0.57}$	66.67$_{\pm 3.40}$
	Sci-ZSEL + Synonym	**80.70**$_{\pm 0.21}$	**86.12**$_{\pm 0.12}$	**98.05**$_{\pm 0.26}$	79.05$_{\pm 3.15}$	**69.74**$_{\pm 1.28}$	**81.21**$_{\pm 2.29}$	**86.21**$_{\pm 1.38}$	**97.45**$_{\pm 0.52}$	**80.95**$_{\pm 1.19}$	**70.26**$_{\pm 4.48}$

Sci-ZSEL + Synonym achieves the highest on all benchmarks while using BLINK, Leads on BC5CDR, QTL_{VT}, and QTL_{LPT}, while Synonym performs best on NCBI and QTL_{CMO} for ReS.

Conclusion

- A scalable, reusable blueprint for zero-shot domain adaptation in specialized, low-resource scientific domains.
- Sci-ZSEL
  - Bounds LLM alias generation to selected entities
  - Employs an ontology-aware filter to discard drifted aliases
- Animal Science Benchmark

Data & Code



Acknowledgement

This work is supported by NSF and USDA grants.





Thank you!